Atlantis on the ZX Spectrum - how the code fits together

repo dalixinc/zx-engine, branch main

Start-up and loader · The main loop · Engine core (shared with Engine Demo 1) · Atlantis game pieces · Actors · Graphics data · Test builds only (not on the tape) · Spectrum hardware

3 = step number in the main loop (0 = once, at start-up) · = this file calls or reads that one · = signal or data from the Spectrum itself · = build step (make)

TV signal
the ULA chip
50 times a second

ROM font
the letter shapes
built into the ROM

Keys + joystick
keyboard rows
Kempston port

Engine core - shared with Engine Demo 1

interrupts.asm
the 50-times-a-second heartbeat: wakes the loop at the start of every frame, counts frames

text.asm
prints words and numbers on screen, without calling the ROM

finds the cell

screen.asm
clear screen, border colour, and the row table: where each of the 192 pixel lines starts

row table

input.asm
reads QAOP, 5678 and the Kempston joystick into one byte, a bit per direction

sprite.asm
shared sprite helpers: set position, pick the right shifted picture

constants.asm
names for numbers: memory addresses, colours, key rows. Used by every file

main.asm

the conductor: calls everything, in order

0. Set-up, once (entry): interrupts, row table, clear the screen, seabed, title text

then, 50 times a second (main_loop):

1. Wait for the next frame
2. Read the keys (and the stats key)
3. Rub out the sprites: diver, bomb, fish
4. Draw them in their new places: fish, bomb, diver
5. Copy the seabed onto the screen (diver after, if on it)
6. Scroll the seabed buffers ready for next frame
7. Put the stats line on top, if switched on
8. Move the diver, fish and bomb (the thinking)

...and back to step 1, for ever

go!

Start-up

loader.asm
a one-line BASIC loader: CLEAR, LOAD "" CODE, then RANDOMIZE USR 32768 jumps to entry at $8000

Actors - the things that move

diver.asm
the player: steers, turns, kicks; stops on rocks and silos

fish.asm
the test fish: swims to and fro, wags its tail

wheel.asm
the Trigon bomb: slides in, bobs, spins, sets its colour

reads keys

asks where they are, which picture

draws with it

position, picture

fastsprite.asm
Atlantis's fast drawing: photograph the water, paste it back, draw. Used for the diver, the fish and the bomb's pixels

reads the pictures

Graphics data

gfx/*.txt
the art, drawn with text characters

gfx2asm.py
in tools/: makes 8 shifted, masked copies

gfx/*.asm
fish, diver and wheel picture bytes

Atlantis game pieces

scenery.asm
paints the seabed colours, then asks scroll for the first picture

first picture

scroll.asm
the seabed: off-screen strips, rocks, silos, weed, copied to the screen

stats.asm
the Symbol Shift + S stats line on the top row

layout.asm
where the seabed rows are, their colours, the scroll buffer addresses

Test builds only

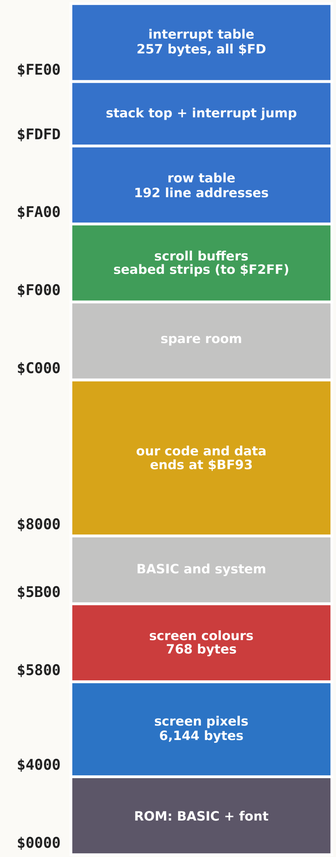
soak.asm
self-test: plays a scripted dive and checks for stray pixels

Switches in main.asm (IFDEF):

TIMING_BARS	border stripes show time used
SLACK_LINES	counts spare scanlines
SHOT_FRAME	freezes for a screenshot
NO_WHEEL	leaves the bomb out

Where things live in memory

48K Spectrum, top of memory first



(not to scale)